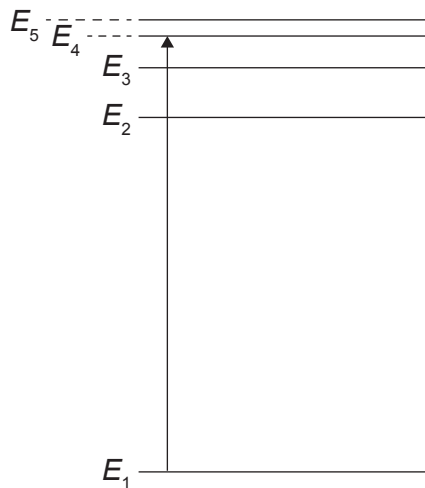


E.1 / PAPER 1 / QUESTION 6

29 April 2025 / TZ3 (file label) / HL Paper 1A / original Q32 / 2225-6719

6. The energy levels of an atom in which an electron transitions from E_1 to E_4 are shown. E_1 is the ground state.

diagram to scale



How many different wavelengths can be emitted when electrons return to the ground state from E_4 ?

- A. 1
- B. 3
- C. 6
- D. 8